

BERLIN, NH, USA

WMO: 726160

Lat: 44.576N

Lon: 71.179W

Elev: 1158

StdP: 14.09

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 94700

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-9.7	-23.6	1.6	-14.4	-17.8	2.2	-8.0	19.7	22.0	17.7	18.3	1.3	290	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.1	85.8	69.3	82.7	67.3	80.6	66.1	71.9	81.5	70.2	79.2	68.4	76.6	6.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.1	111.5	76.4	66.8	103.0	74.4	65.3	97.7	73.0	36.2	81.2	34.8	79.3	33.3	76.9	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.9	14.6	12.4	-25.4	90.5	6.1	2.6	-29.8	92.4	-33.4	93.9	-36.8	95.3	-41.3	97.2		
			-24.9	74.6	5.0	1.6	-28.5	75.7	-31.5	76.6	-34.3	77.5	-37.9	78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	14.7	17.4	26.2	39.6	51.8	60.6	65.3	63.9	56.9	44.6	33.3	21.6
	DBStd	19.92	12.83	11.67	11.09	8.13	7.67	6.56	5.40	5.39	7.49	8.08	9.29	11.18
	HDD50	4764	1095	914	740	329	69	3	0	0	20	210	503	881
	HDD65	8778	1560	1334	1203	762	415	165	63	87	259	634	950	1346
	CDD50	1641	0	0	2	17	125	320	476	430	227	41	3	0
	CDD65	180	0	0	0	0	5	32	74	52	17	0	0	0
	CDH74	2644	0	0	3	23	185	510	971	676	262	14	0	0
	CDH80	627	0	0	0	3	44	130	248	142	60	0	0	0
Wind	WSAvg	4.0	4.3	4.5	5.0	5.2	4.5	3.6	3.0	2.6	2.9	3.9	4.0	3.9
Precipitation	PrecAvg	37.4	2.4	2.2	2.4	3.0	3.3	3.7	3.4	4.0	3.1	3.4	3.8	3.0
	PrecMax	53.8	7.0	6.5	5.2	5.6	8.5	6.6	5.4	7.8	7.0	6.2	9.4	8.8
	PrecMin	30.9	0.2	0.2	0.4	1.3	0.9	1.5	1.4	1.0	0.7	1.3	1.3	0.9
	PrecStd	5.2	1.7	1.3	1.0	1.4	1.7	1.5	1.1	1.7	1.6	1.5	1.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	50.2	63.3	77.1	85.2	88.9	89.3	88.0	86.5	75.8	64.1	51.6
		MCWB	46.0	42.5	52.5	58.1	64.9	70.1	72.6	70.9	70.1	61.4	55.8	46.4
	2%	DB	42.6	43.4	53.1	69.6	79.5	83.8	85.6	83.6	80.9	69.9	58.7	45.0
		MCWB	38.7	37.5	43.4	53.1	60.2	67.0	70.1	68.5	66.6	57.8	51.3	41.6
	5%	DB	36.7	38.1	46.5	62.7	74.0	80.5	82.5	81.0	76.6	64.0	53.4	39.8
		MCWB	33.9	34.0	39.3	49.4	57.8	65.3	68.1	67.1	64.6	55.5	47.7	37.0
	10%	DB	33.7	34.9	42.5	56.0	69.7	76.0	80.0	78.1	72.2	59.4	47.5	36.4
		MCWB	31.4	31.9	36.3	45.1	56.0	62.5	66.6	65.5	62.1	53.3	42.9	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	45.1	53.3	60.7	67.2	72.5	75.0	73.3	71.3	65.1	58.7	47.7
		MCDB	49.4	49.3	62.3	72.4	79.7	83.1	84.9	81.8	82.0	71.7	62.5	49.6
	2%	WB	38.8	38.6	45.2	55.8	64.0	70.4	72.5	71.0	68.7	60.2	53.2	41.7
		MCDB	41.8	42.5	49.6	67.1	74.2	80.4	81.8	79.8	77.8	66.6	57.3	44.2
	5%	WB	34.4	34.8	40.1	50.8	60.8	67.8	70.5	69.2	66.4	57.0	47.5	37.6
		MCDB	36.4	37.5	45.9	59.5	70.0	76.0	79.3	77.0	73.7	62.9	51.8	39.5
	10%	WB	31.1	32.2	36.8	46.5	58.0	65.1	68.4	67.4	63.8	53.7	43.7	34.1
		MCDB	33.6	34.9	41.6	54.7	66.2	72.6	76.5	74.9	70.0	58.6	47.2	36.0
Mean Daily Temperature Range	5% DB	MDBR	21.6	23.5	22.1	23.1	26.1	24.8	25.1	25.3	25.4	21.9	18.2	18.2
		MCDBR	23.0	25.0	27.6	37.6	36.7	31.5	29.2	28.8	30.7	31.3	27.3	20.0
		MCWBR	20.2	20.3	19.3	22.0	19.5	16.2	15.0	15.1	17.6	20.5	20.4	17.7
	5% WB	MCDBR	22.6	22.0	24.0	32.4	29.0	24.7	25.3	24.6	26.2	26.5	23.2	18.9
		MCWBR	20.3	18.5	17.7	21.1	16.7	13.9	13.6	13.7	15.6	18.6	18.8	17.4
Clear-Sky Solar Irradiance	taub	0.253	0.274	0.308	0.341	0.363	0.407	0.415	0.390	0.355	0.319	0.301	0.266	
	taud	2.421	2.332	2.352	2.406	2.406	2.276	2.261	2.329	2.423	2.531	2.517	2.453	
	Ebn at Noon	281	292	295	292	287	274	270	273	275	273	260	264	
	Edh at Noon	23	30	34	35	36	42	42	37	31	25	21	20	
All-Sky Solar Radiation	RadAvg	468	745	1119	1381	1558	1702	1726	1566	1279	770	485	356	
	RadStd	53	81	71	161	196	145	147	93	89	77	58	30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	N/A

Nomenclature: See separate page